Alexander (Sasha) Apartsin

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SELECTED RECOGNITION & ACHIEVEMENTS

AI Product of the Year recognition for SafeRide automotive AI product work.

$250K Citi LabLink grant for a financial data science collaboration platform.

IIA innovation grants for Motorola Solutions industry-academia AI projects.

$10M hardware-cost savings contribution through model-based capacity planning at Amdocs.

INNOVATION & ENTREPRENEURSHIP

Founder / CTO experience across search, multimedia, cloud optimization, and applied AI ventures; corporate innovation through AI labs, grants, patents, and prototypes.

RESEARCH & IMPACT

Publications: 40+ Scholar-indexed works and 470+ citations across ML, vision, NLP, deep learning, and signal processing.

Patents: 10 granted/filed across vehicle diagnostics, driver monitoring, model configuration, onboard inference, public-safety audio, and search/image matching.

TEACHING & MENTORSHIP

24+ supervised B.Sc. / M.Sc. final projects since 2012, with teaching across HIT, Afeka, TAU, and BIU; multi-course, project-based series in Language, Vision, and Scalable AI, each with a companion online textbook, delivered over 16 past offerings with 112+ student course projects.

DOMAINS

Vision | Audio | Speech | Text / NLP | LLMs | Diffusion | Task Allocation | Anomaly Detection | RecSys | Time Series / Signals | Novel Material Discovery | Big Data

INDUSTRIES

Automotive | Finance | Public Safety | Healthcare | Telecom | Media / Multimedia | Document AI | Education | Search | Industrial Inspection

HANDS-ON

Python | PyTorch | Hugging Face | Spark NLP / spaCy | OpenCV | LangChain / LangGraph | OpenAI API | RAG | C / C++ | SQL | Pandas / NumPy | Databricks | MLflow | Horovod

- **AI researcher and technology leader** with 30+ years of experience across machine learning, signal processing, computer vision, NLP, and generative AI.
- **End to end, from research to product:** builds and leads research and engineering teams, drives cross-functional collaboration, co-defines products with customers, and executes delivery from original research and algorithms through production systems to shipped products, staying hands-on alongside the team.
- **Academic and applied impact** through research, teaching, and student supervision, with applications across healthcare, automotive, public safety, finance, and enterprise AI.

LEADERSHIP & RESEARCH EXPERIENCE

Faculty Member - HIT, School of Computer Science — **2024-Present**

Conducts applied AI research and supervises student projects; teaches graduate AI courses at Bar-Ilan on LLMs, GenAI, and agents; created undergraduate Hands-On AI courses in LLMs and GenAI.

Head of Data Science - Questar / SafeRide Technologies — **2020-2024**

Built and hired the data science team; led big-data AI for vehicle telematics across signal, audio, graph, anomaly detection, and onboard-model work; led OEM-partner projects including engine-health monitoring (Ford), steering-by-wire monitoring (Hyundai Mobis), in-cabin acoustic fault detection (Renault), fuel-consumption modeling (Aptiv), and fault-sequence analysis (M.A.N.); engaged directly with customers to shape requirements; led the definition of a no-code product for onboard monitoring models.

VP of Artificial Intelligence - harmon.ie — **2019-2020**

Led AI for an enterprise document-management product designed to help organizations handle the growing amount of information in Office 365; managed a cross-functional data-science and engineering team and shaped the research agenda across large-scale text understanding, user-behavior modeling from Office-graph signals, and recommendation under limited supervision.

Head of AI Research - Motorola Solutions, Israel Innovation Center — **2015-2019**

Built and led the research team at Motorola Solutions' first AI lab in the Innovation Center; designed and led execution of the roadmap across video, text, speech, and public-safety AI; established internal cross-functional partnerships and academia-industry collaborations funded by Israel Innovation Authority programs and binational grants.

AVP Data Science - Citi Israel Innovation Lab — **2013-2015**

Led financial data-science projects: bond-price prediction for auctions, FX causality analysis and trader-informativeness classification, anomaly detection for trader compliance auditing, and trade-to-wire matching via correlation and constraint optimization.

Founder / CTO / R&D and algorithm leadership - SearchNG, CloudTuner, MPCode, Tegrity, AppiVision, Amdocs, Comverse, earlier roles — **1991-2012**

Founded ventures and led hands-on intelligent-system work in search, multimedia, video streaming, wireless communications, telecom, predictive modeling, handwriting recognition, imaging, inspection, and signal acquisition.

EDUCATION

Postdoctoral Fellowship - Weizmann Institute of Science · computer vision and machine learning — **2013**

Ph.D., Computer Science - Tel Aviv University · machine learning and signal processing — **2012**

M.Sc., Technology Management - Polytechnic Institute of NYU — **2006**

M.Sc., Computer Science - Weizmann Institute of Science · computer vision — **2004**

B.Sc., Computer Science - Technion — **1995**